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(54) **DRIVING ASSISTANCE DEVICE, DRIVING ASSISTANCE METHOD, AND STORAGE MEDIUM**

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(Continued)

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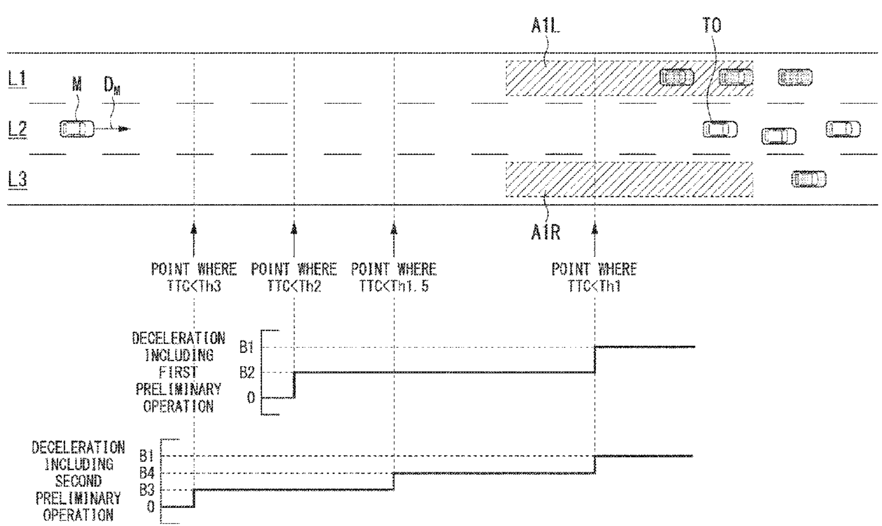
(74) *Attorney, Agent, or Firm* — Amin, Turocy & Watson, LLP

(57) **ABSTRACT**

At least one processor executes one or both of a braking control process of stopping a vehicle, and a steering-based avoidance control process of avoiding a collision with the object, execute a first preliminary operation control process of performing a first preliminary operation of notifying a driver that the object is present; and execute a second preliminary operation control process of performing a second preliminary operation of notifying the driver that the object is present when the indicator value is less than a third threshold and there is no travel path along which the vehicle is able to travel after the vehicle avoids the collision with the object in the steering. Control content is changed in accordance with a type of the object in at least some of the steering-based avoidance control process, the first preliminary operation control process, and the second preliminary operation control process.

10 Claims, 10 Drawing Sheets



(51) **Int. Cl.**
B60W 10/20 (2006.01)
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(52) **U.S. Cl.**
CPC *B60W 2554/402* (2020.02); *B60W 2554/4026* (2020.02); *B60W 2710/18* (2013.01); *B60W 2710/20* (2013.01)

(58) **Field of Classification Search**
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USPC 701/26
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FIG. 1

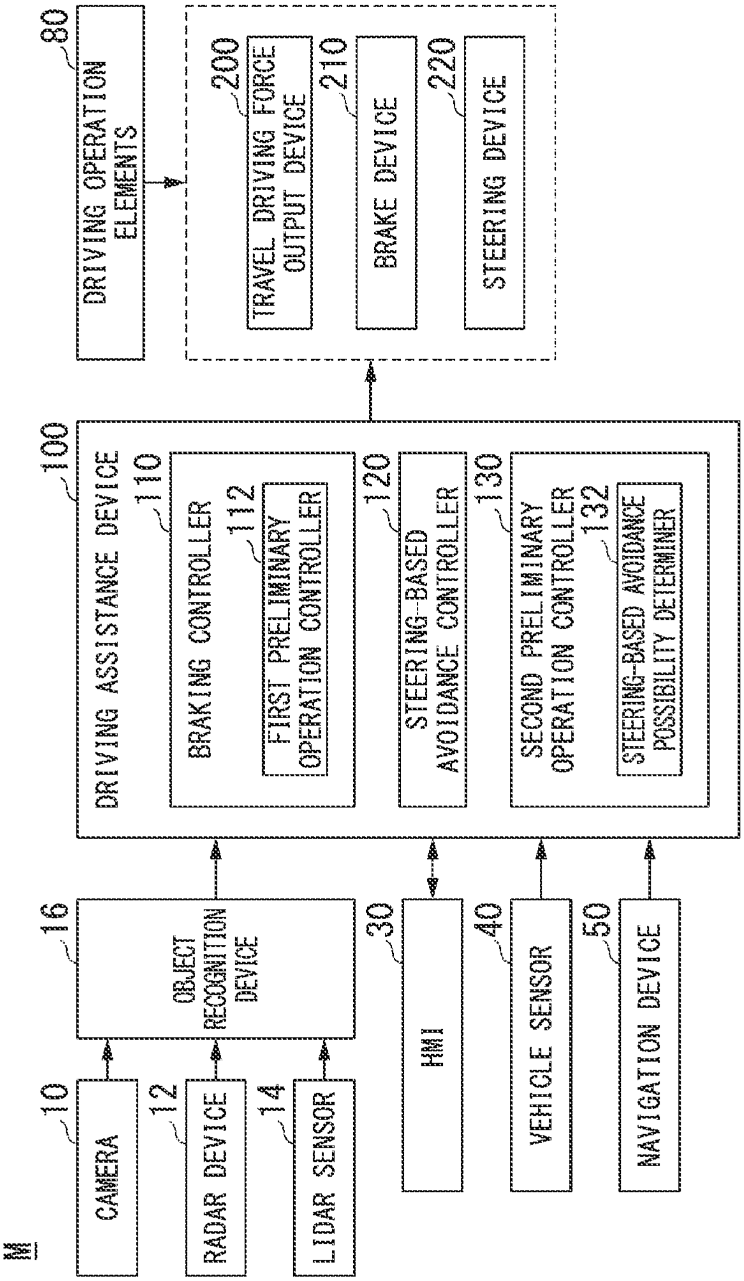

M
CAMERA
10
RADAR DEVICE
12
LIDAR SENSOR
14
OBJECT RECOGNITION DEVICE
16
HMI
30
VEHICLE SENSOR
40
NAVIGATION DEVICE
50
DRIVING ASSISTANCE DEVICE
100
BRAKING CONTROLLER
110
FIRST PRELIMINARY OPERATION CONTROLLER
112
STEERING-BASED AVOIDANCE CONTROLLER
120
SECOND PRELIMINARY OPERATION CONTROLLER
130
STEERING-BASED AVOIDANCE POSSIBILITY DETERMINER
132
DRIVING OPERATION ELEMENTS
80
TRAVEL DRIVING FORCE OUTPUT DEVICE
200
BRAKE DEVICE
210
STEERING DEVICE
220

FIG. 2

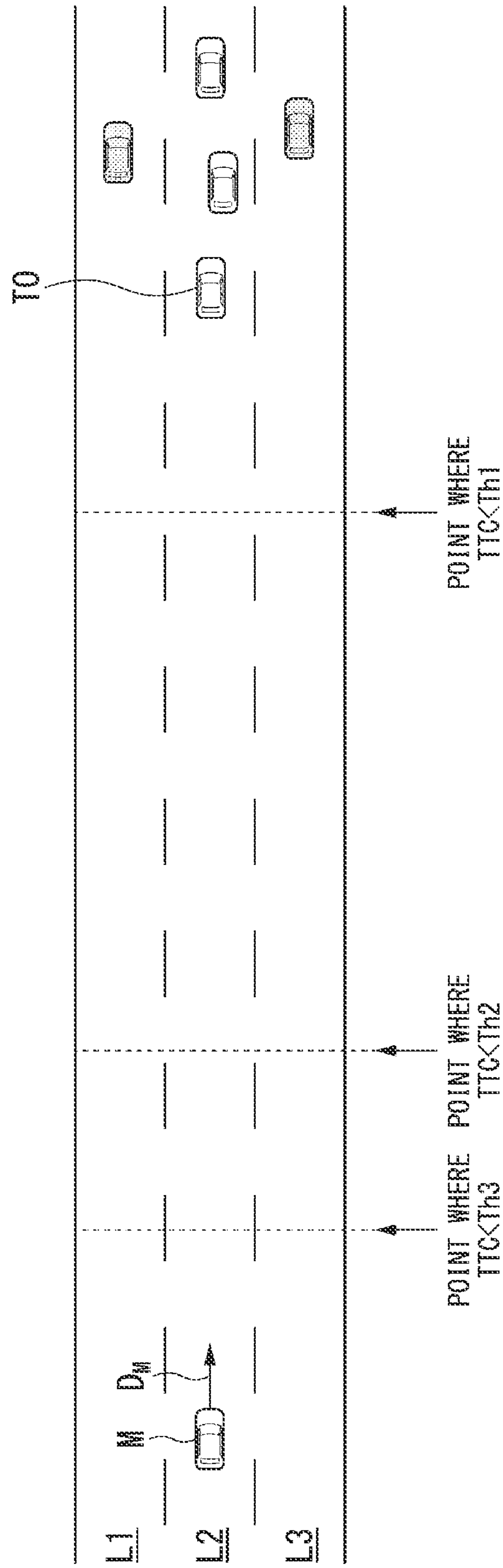

TO
POINT WHERE TTC<Th1
POINT WHERE TTC<Th2
POINT WHERE TTC<Th3
M
D_M
L1
L2
L3

FIG. 3

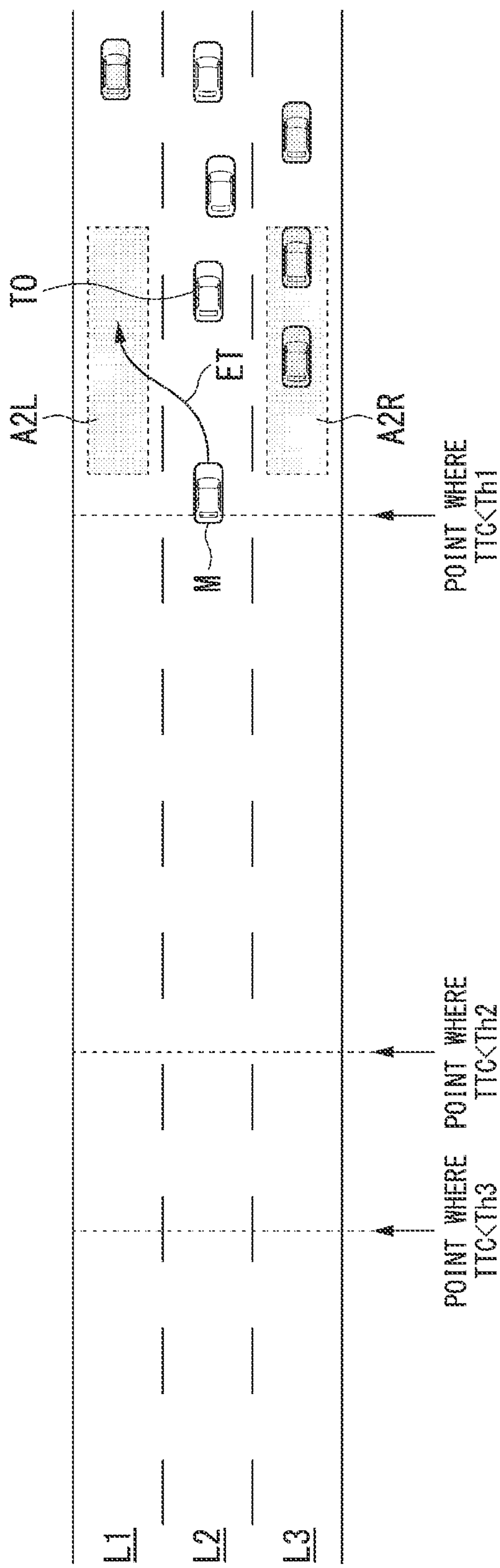

TO
ET
A2L
A2R
M
POINT WHERE TTC<Th1
POINT WHERE TTC<Th2
POINT WHERE TTC<Th3
L1
L2
L3

FIG. 4

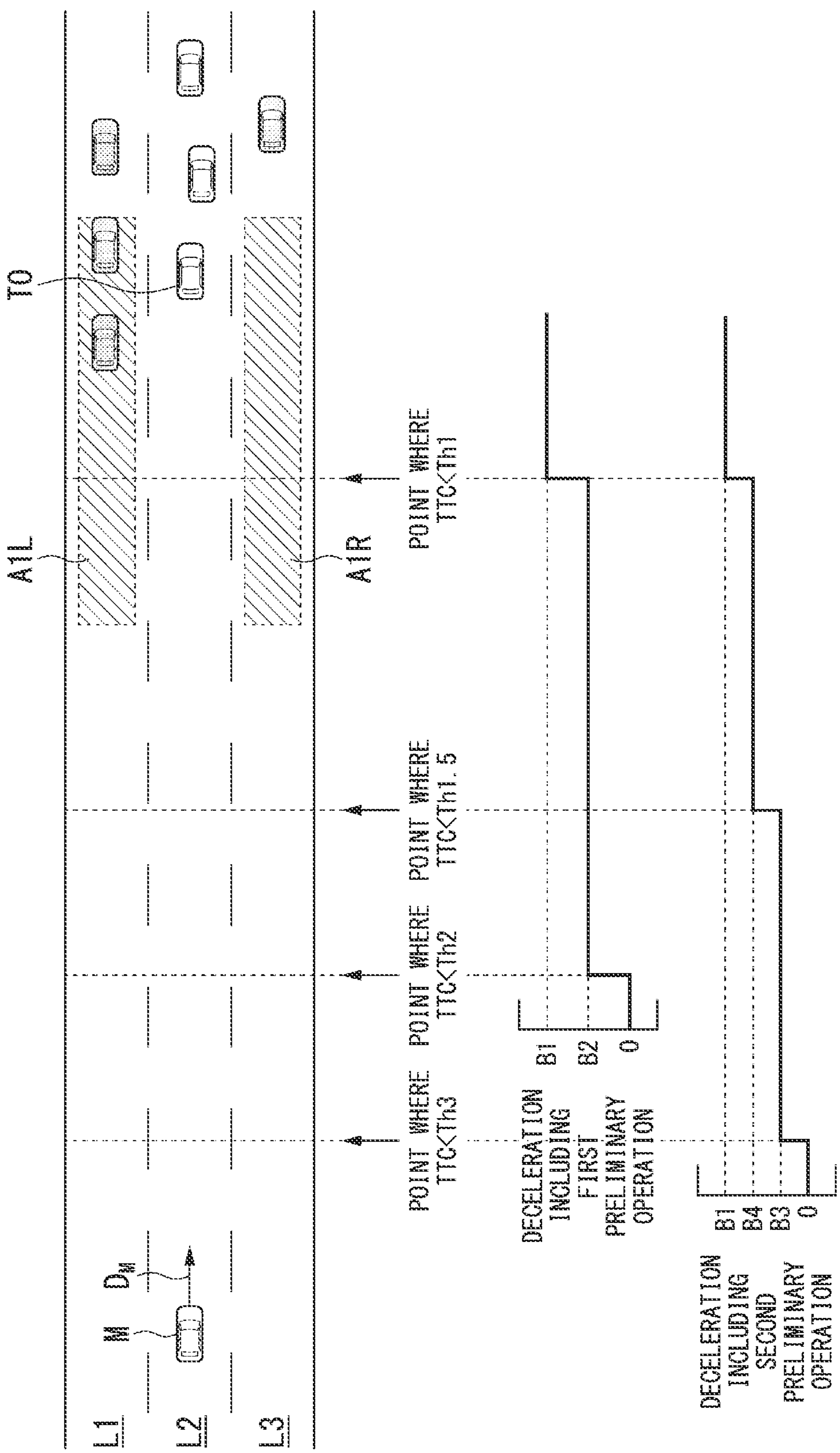

L1
L2
L3
M
D_M
T0
A1L
A1R
POINT WHERE TTC<Th3
POINT WHERE TTC<Th2
POINT WHERE TTC<Th1.5
POINT WHERE TTC<Th1
DECELERATION INCLUDING FIRST PRELIMINARY OPERATION
B1
B2
0
DECELERATION INCLUDING SECOND PRELIMINARY OPERATION
B1
B4
B3
0

FIG. 5

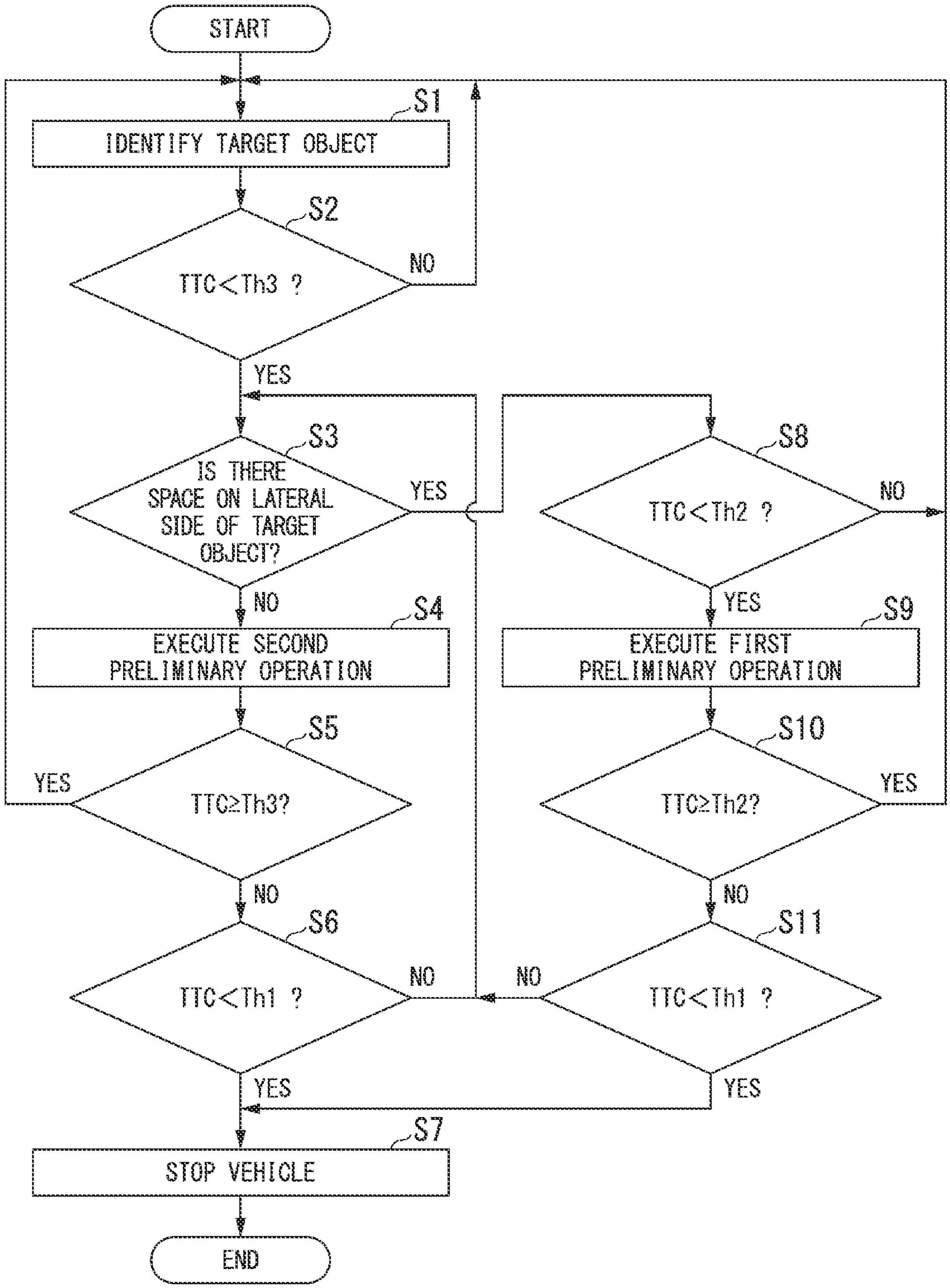
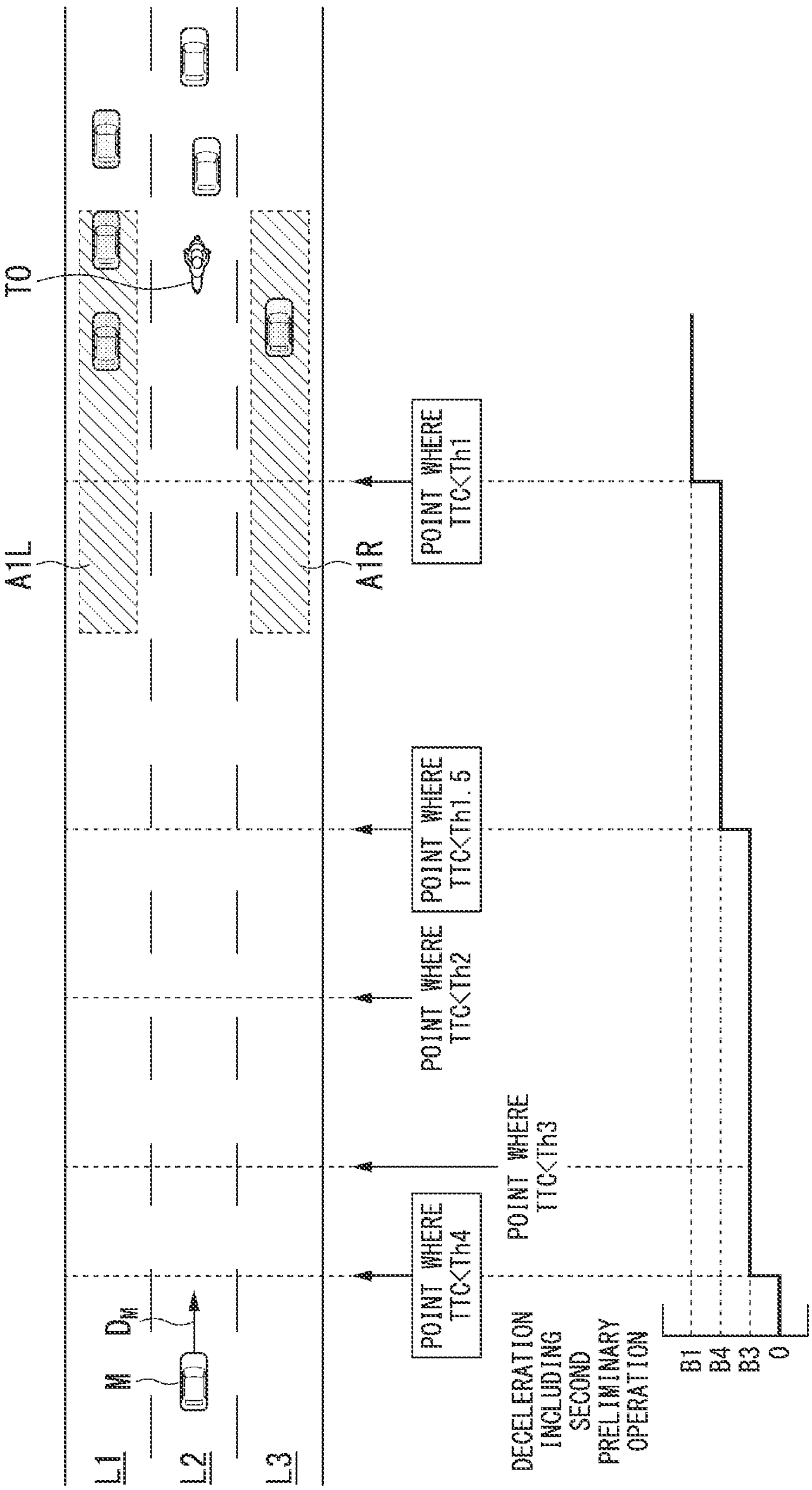
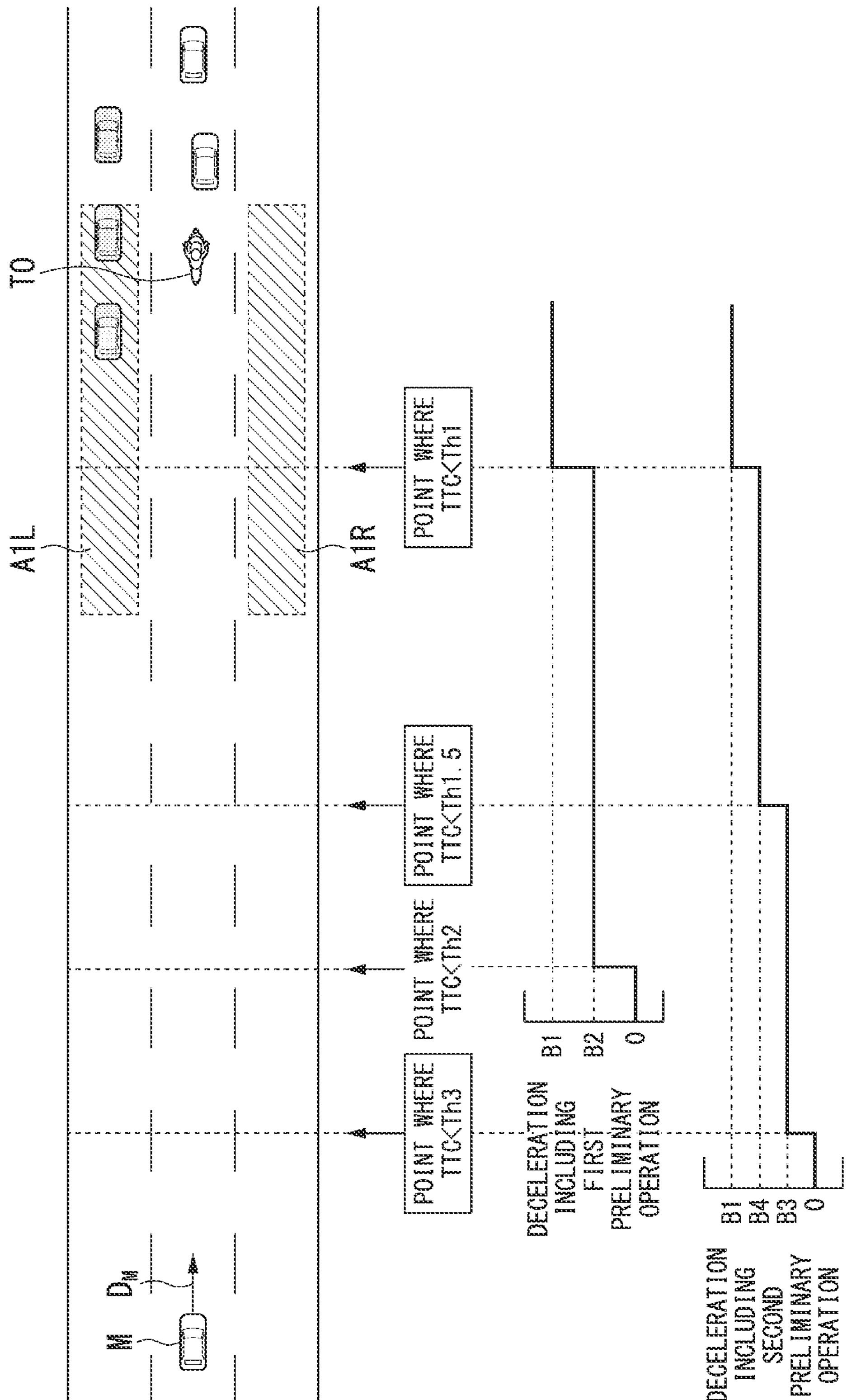

START
S1
IDENTIFY TARGET OBJECT
S2
TTC<Th3 ?
NO
YES
S3
IS THERE SPACE ON LATERAL SIDE OF TARGET OBJECT?
YES
NO
S4
EXECUTE SECOND PRELIMINARY OPERATION
S5
TTC≥Th3?
YES
NO
S6
TTC<Th1 ?
NO
YES
S7
STOP VEHICLE
END
S8
TTC<Th2 ?
NO
YES
S9
EXECUTE FIRST PRELIMINARY OPERATION
S10
TTC≥Th2?
YES
NO
S11
TTC<Th1 ?
NO
YES

FIG. 6



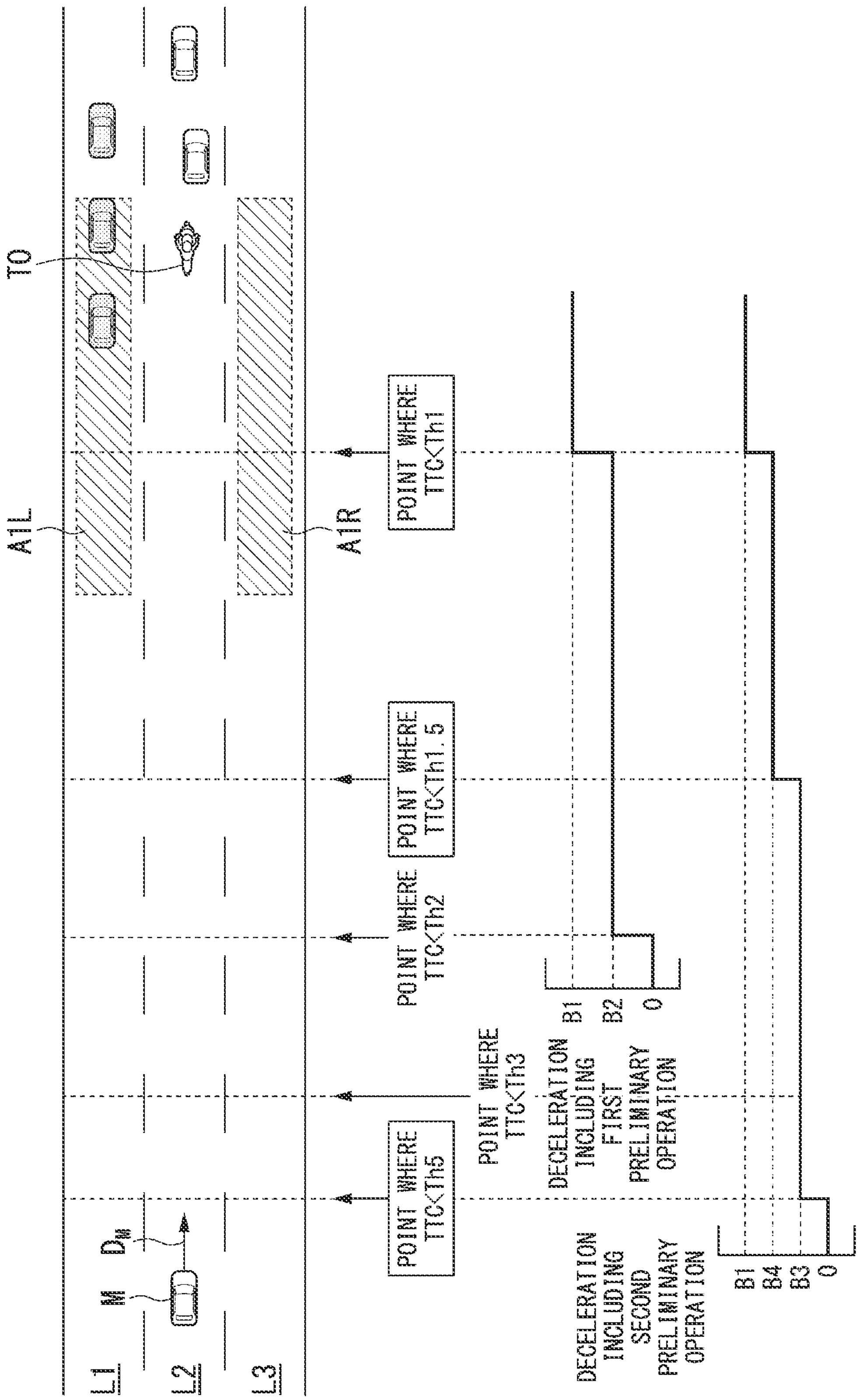
T0
A1L
A1R
POINT WHERE TTC<Th1
POINT WHERE TTC<Th1.5
POINT WHERE TTC<Th2
POINT WHERE TTC<Th3
POINT WHERE TTC<Th4
M
D_M
L1
L2
L3
DECELERATION INCLUDING SECOND PRELIMINARY OPERATION
B1
B4
B3
0

FIG. 7



L1
L2
L3
M
D_M
T0
A1L
A1R
POINT WHERE TTC<Th3
POINT WHERE TTC<Th2
POINT WHERE TTC<Th1.5
POINT WHERE TTC<Th1
DECELERATION INCLUDING FIRST PRELIMINARY OPERATION
B1
B2
0
DECELERATION INCLUDING SECOND PRELIMINARY OPERATION
B1
B4
B3
0

FIG. 8



L1
L2
L3
M
D_M
T0
A1L
A1R
POINT WHERE TTC<Th1
POINT WHERE TTC<Th1.5
POINT WHERE TTC<Th2
POINT WHERE TTC<Th3
POINT WHERE TTC<Th5
DECELERATION INCLUDING FIRST PRELIMINARY OPERATION
B1
B2
0
DECELERATION INCLUDING SECOND PRELIMINARY OPERATION
B1
B4
B3
0

FIG. 9

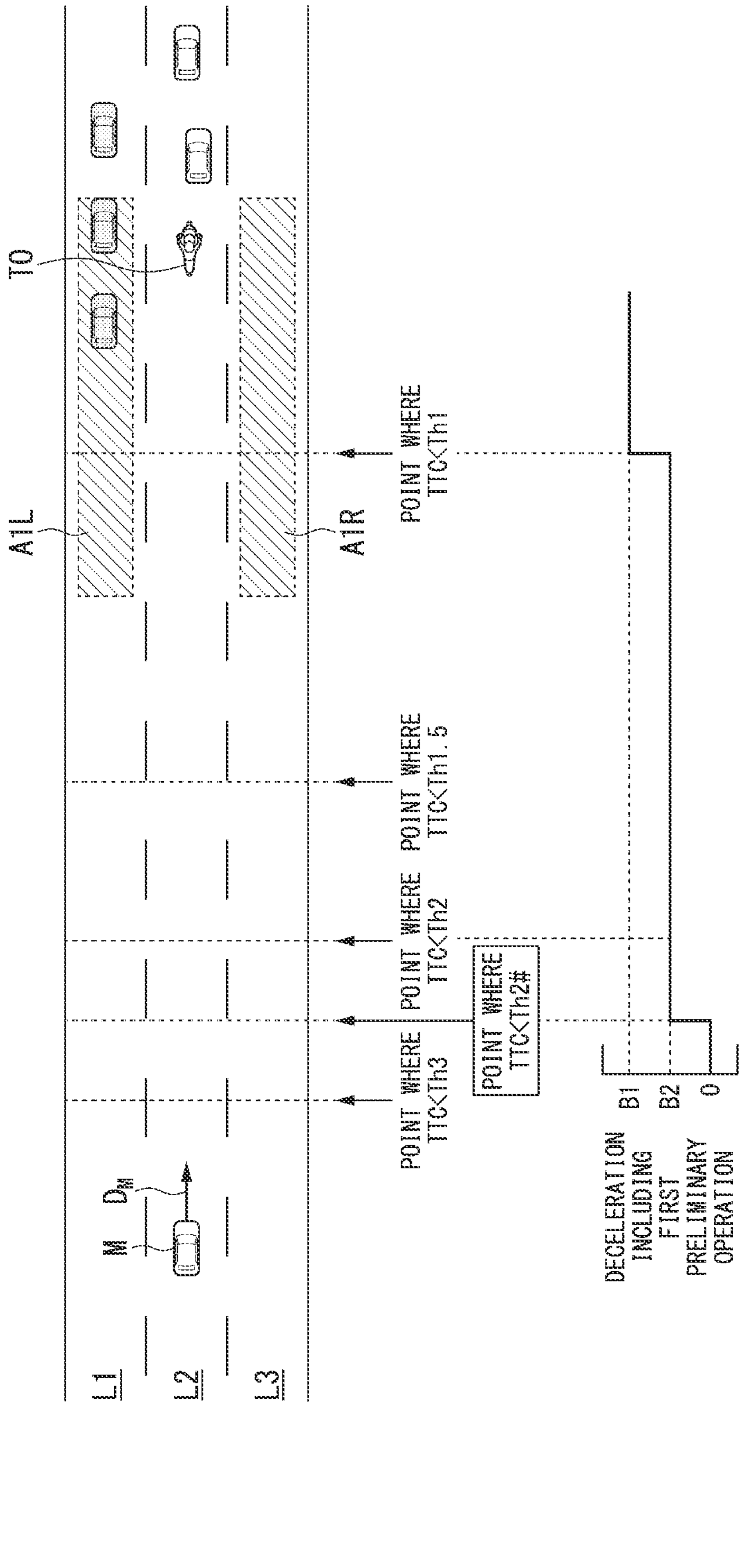

T0
A1L
A1R
M
D_M
L1
L2
L3
POINT WHERE TTC<Th1
POINT WHERE TTC<Th1.5
POINT WHERE TTC<Th2
POINT WHERE TTC<Th3
POINT WHERE TTC<Th2#
B1
B2
0
DECELERATION INCLUDING FIRST PRELIMINARY OPERATION

FIG. 10

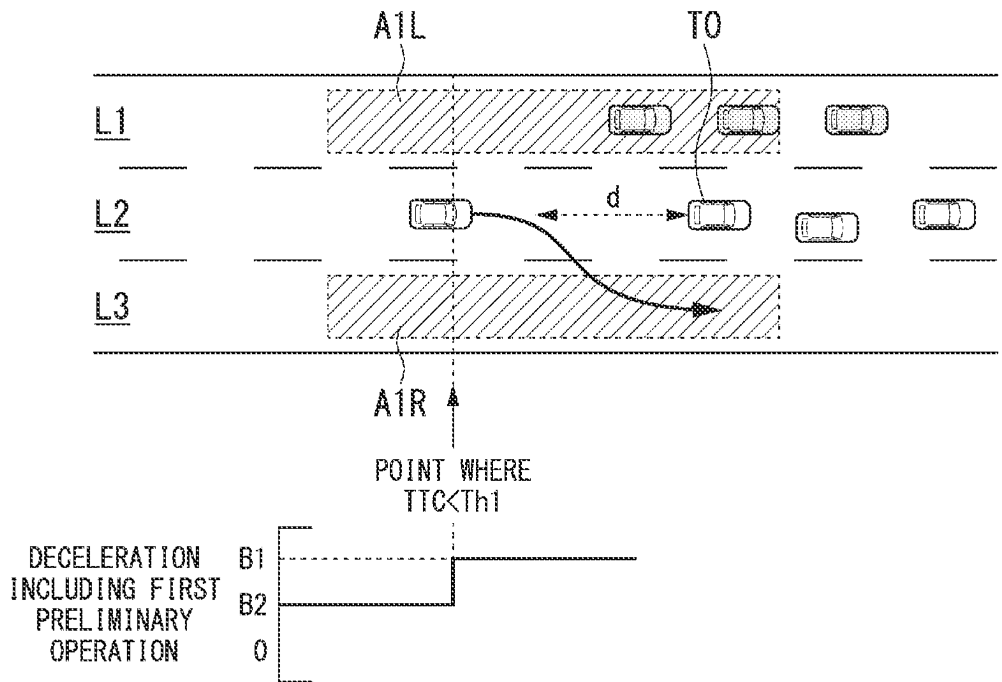
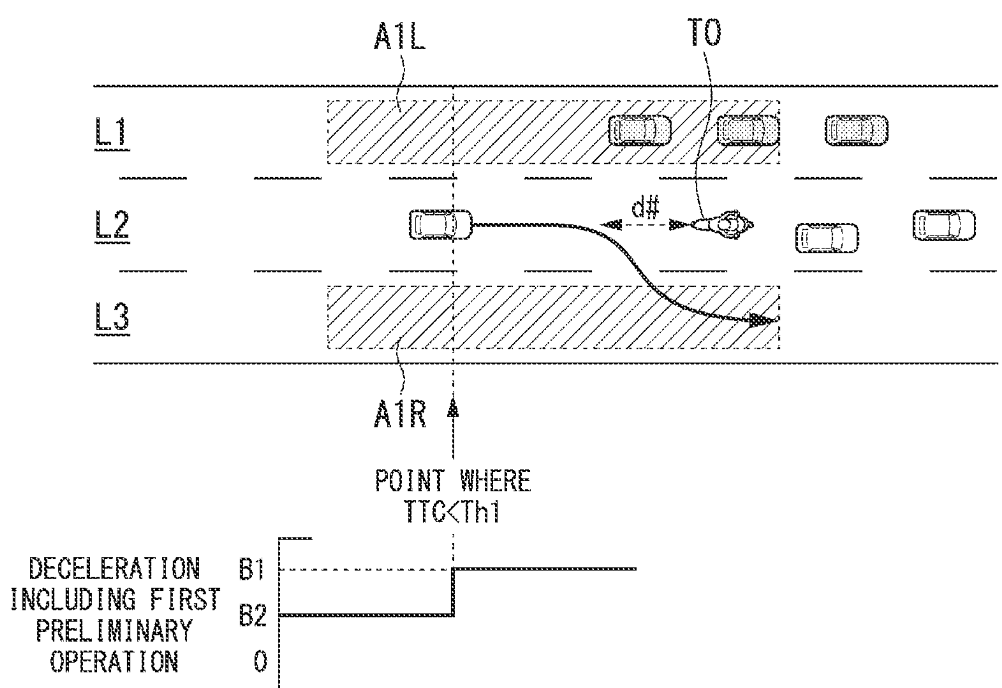

A1L
TO
L1
L2
d
L3
A1R
POINT WHERE
TTC<Th1
DECELERATION
INCLUDING FIRST
PRELIMINARY
OPERATION
B1
B2
0

FIG. 11


A1L
TO
L1
L2
d#
L3
A1R
POINT WHERE
TTC<Th1
DECELERATION
INCLUDING FIRST
PRELIMINARY
OPERATION
B1
B2
0

DRIVING ASSISTANCE DEVICE, DRIVING ASSISTANCE METHOD, AND STORAGE MEDIUM

CROSS-REFERENCE TO RELATED APPLICATION

Priority is claimed on Japanese Patent Application No. 2022-043717, filed Mar. 18, 2022, the content of which is incorporated herein by reference.

BACKGROUND

Field of the Invention

The present invention relates to a driving assistance device, a driving assistance method, and a storage medium.

Description of Related Art

Recently, an invention of a vehicle control device for performing automated deceleration control and automated steering control has been disclosed (Japanese Unexamined Patent Application, First Publication No. 2020-50010).

SUMMARY

In a vehicle capable of performing automated steering control in addition to automated deceleration control, a probability that any sudden change in a surrounding environment of a vehicle can be coped with quickly becomes high and a degree of control margin becomes relatively high. On the other hand, because automated steering control becomes difficult if there is no avoidance space on a lateral side of a target object, a degree of control margin is no different from that of a vehicle that performs only automated deceleration control. In the conventional technology, it may be difficult to perform an operation corresponding to the above-described environment difference.

The present invention has been made in consideration of such circumstances and an objective of the present invention is to provide a driving assistance device, a driving assistance method, and a storage medium capable of performing an appropriate operation corresponding to a surrounding situation of a target object.

A driving assistance device, a driving assistance method, and a storage medium adopt the following configurations.

(1): According to an aspect of the present invention, there is provided a driving assistance device including: a storage medium storing computer-readable instructions; and at least one processor connected to the storage medium, the at least one processor executing the computer-readable instructions to: execute a referring process of referring to an output of a detection device for detecting that an object is located in front of a vehicle; execute one or both of a braking control process or a steering based avoidance control process; the braking control process being a process that instructs a brake device of the vehicle to stop the vehicle when an indicator value that decreases as the vehicle approaches the object is less than a first threshold, the a steering-based avoidance control process being a process that instruct a steering device of the vehicle to avoid a collision with the object in steering, execute a first preliminary operation control process of performing a first preliminary operation of notifying a driver of the vehicle that the object is present when the indicator value is less than a second threshold; and execute a second preliminary operation control process of performing a second preliminary operation of notifying the driver of the vehicle that the object is present when the indicator value is less than a third threshold and it is determined, at a time point when the indicator value is less than the third threshold, that there is no travel path along which the vehicle is able to travel on both left and right sides of the object after the vehicle avoids the collision with the object in the steering, wherein the first threshold is less than the second threshold and the second threshold is less than the third threshold, and wherein control content is changed in accordance with a type of the object in at least some of the steering-based avoidance control process, the first preliminary operation control process, and the second preliminary operation control process.

(2): In the above-described aspect (1), the at least one processor changes control content in accordance with whether the object is a four-wheeled vehicle or a two-wheeled vehicle in at least some of the steering-based avoidance control process, the first preliminary operation control process, and the second preliminary operation control process.

(3): In the above-described aspect (1) or (2), when the object is a two-wheeled vehicle, the driving assistance device suppresses proximity of the vehicle to the two-wheeled vehicle by actively operating the brake device of the vehicle.

(4): In any one of the above-described aspects (1) to (3), the at least one processor suppresses the avoidance based on the steering in the steering-based avoidance control process when the object is a two-wheeled vehicle as compared with when the object is a four-wheeled vehicle.

(5): In the above-described aspect (4), the at least one processor performs the second preliminary operation at a time point after the indicator value is less than a fourth threshold when the indicator value is less than the fourth threshold, it is determined, at the time point when the indicator value is less than the fourth threshold, that there is no travel path along which the vehicle is able to travel on both left and right sides of the object after the vehicle avoids the collision with the object in the steering, and the object is a four-wheeled vehicle, the at least one processor performs the second preliminary operation at a time point when the indicator value is less than the fourth threshold when the indicator value is less than the fourth threshold, it is determined, at a time point when the indicator value is less than the fourth threshold, that there is no travel path along which the vehicle is able to travel on both left and right sides of the object after the vehicle avoids the collision with the object in the steering, and the object is a two-wheeled vehicle, and the third threshold is less than the fourth threshold.

(6): In any one of the above-described aspects (1) to (5), the at least one processor does not perform the second preliminary operation at a time point when the indicator value between a four-wheeled vehicle and the vehicle is less than the third threshold when the indicator value is less than the third threshold, it is determined, at a time point when the indicator value is less than the third threshold, that there is a travel path along which the vehicle is able to travel on one of the left and right sides of the object after the vehicle avoids the collision with the object in the steering, and the object is the four-wheeled vehicle, and performs the first preliminary operation when the indicator value between the four-wheeled vehicle and the vehicle is less than the second threshold, and the at least one processor performs the second preliminary operation at a time point when the indicator value between a two-wheeled vehicle and the vehicle is less than the third threshold when the indicator value is less than the third threshold, it is determined, at the time point when the indicator value is less than the third threshold, that there is a travel path along which the vehicle is able to travel on one of the left and right sides of the object after the vehicle avoids the collision with the object in the steering, and the object is the two-wheeled vehicle.

(7): In any one of the above-described aspects (1) to (6), the at least one processor does not perform the second preliminary operation at a time point when the indicator value is less than a fifth threshold when the indicator value is less than the fifth threshold, it is determined, at the time point when the indicator value is less than the fifth threshold, that there is a travel path along which the vehicle is able to travel on one of the left and right sides of the object after the vehicle avoids the collision with the object in the steering, and the object is a four-wheeled vehicle, and performs the first preliminary operation when the indicator value between the four-wheeled vehicle and the vehicle is less than the second threshold, the at least one processor performs the second preliminary operation at a time point when the indicator value between a two-wheeled vehicle and the vehicle is less than the fifth threshold when the indicator value is less than the fifth threshold, it is determined, at the time point when the indicator value is less than the fifth threshold, that there is a travel path along which the vehicle is able to travel on one of the left and right sides of the object after the vehicle avoids the collision with the object in the steering, and the object is the two-wheeled vehicle, the second threshold is less than the fifth threshold, and the third threshold is less than the fifth threshold.

(8): In any one of the above-described aspects (1) to (7), the at least one processor does not perform the first preliminary operation at a time point when the indicator value between a four-wheeled vehicle and the vehicle is less than a sixth threshold when the indicator value is less than the sixth threshold, it is determined, at a time point when the indicator value is less than the sixth threshold, that there is a travel path along which the vehicle is able to travel on one of the left and right sides of the object after the vehicle avoids the collision with the object in the steering, and the object is the four-wheeled vehicle, and performs the first preliminary operation when the indicator value between the four-wheeled vehicle and the vehicle is less than the second threshold, the at least one processor performs the first preliminary operation at a time point when the indicator value between a two-wheeled vehicle and the vehicle is less than the sixth threshold when the indicator value is less than the sixth threshold, it is determined, at the time point when the indicator value is less than the sixth threshold, that there is a travel path along which the vehicle is able to travel on one of the left and right sides of the object after the vehicle avoids the collision with the object in the steering, and the object is the two-wheeled vehicle, and the second threshold is less than the sixth threshold.

(9): According to an aspect of the present invention, there is provided a driving assistance method including: executing, by a driving assistance device, referring process of referring to an output of a detection device for detecting that an object is located in front of a vehicle, one or both of a braking control process of instructing a brake device of the vehicle to stop the vehicle when an indicator value that decreases as the vehicle approaches the object is less than a first threshold, and a steering-based avoidance control process of instructing a steering device of the vehicle to avoid a collision with the object in steering, a first preliminary operation control process of performing a first preliminary operation of notifying a driver of the vehicle that the object is present when the indicator value is less than a second threshold; and executing, by the driving assistance device, a second preliminary operation control process of performing a second preliminary operation of notifying the driver of the vehicle that the object is present when the indicator value is less than a third threshold and it is determined, at a time point when the indicator value is less than the third threshold, that there is no travel path along which the vehicle is able to travel on both left and right sides of the object after the vehicle avoids the collision with the object in the steering, wherein the first threshold is less than the second threshold and the second threshold is less than the third threshold, and wherein control content is changed in accordance with a type of the object in at least some of the steering-based avoidance control process, the first preliminary operation control process, and the second preliminary operation control process.

(10): According to an aspect of the present invention, there is provided a non-transitory computer storage medium storing a program for causing a computer to: execute a referring process of referring to an output of a detection device for detecting that an object is located in front of a vehicle, execute one or both of a braking control process of instructing a brake device of the vehicle to stop the vehicle when an indicator value that decreases as the vehicle approaches the object is less than a first threshold, and a steering-based avoidance control process of instructing a steering device of the vehicle to avoid a collision with the object in steering; execute a first preliminary operation control process of performing a first preliminary operation of notifying a driver of the vehicle that the object is present when the indicator value is less than a second threshold; and execute a second preliminary operation control process of performing a second preliminary operation of notifying the driver of the vehicle that the object is present when the indicator value is less than a third threshold and it is determined, at a time point when the indicator value is less than the third threshold, that there is no travel path along which the vehicle is able to travel on both left and right sides of the object after the vehicle avoids the collision with the object in the steering, wherein the first threshold is less than the second threshold and the second threshold is less than the third threshold, and wherein control content is changed in accordance with a type of the object in at least some of the steering-based avoidance control process, the first preliminary operation control process, and the second preliminary operation control process.

According to the above-described aspects, it is possible to perform an appropriate operation corresponding to a surrounding situation of a target object.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a configuration diagram of a vehicle in which a driving assistance device according to an embodiment is mounted.

FIG. **2** is a diagram showing an overview of a function of the driving assistance device.

FIG. **3** is a diagram showing an example of an operation scene of a steering-based avoidance controller.

FIG. **4** is a diagram for describing a preliminary operation.

FIG. **5** is a flowchart showing an example of a flow of a process executed by the driving assistance device.

FIG. **6** is a diagram (1) for describing a change example of control content.

FIG. **7** is a diagram (2) for describing a change example of control content.

FIG. **8** is a diagram (3) for describing a change example of control content.

FIG. **9** is a diagram (4) for describing a change example of control content.

FIG. **10** is a diagram (5) for describing a change example of control content.

FIG. **11** is a diagram (6) for describing a change example of control content.

DESCRIPTION OF EMBODIMENTS

Hereinafter, embodiments of a driving assistance device, a driving assistance method, and a storage medium according to the present invention will be described with reference to the drawings. As used throughout this disclosure, the singular forms "a," "an," and "the" include plural reference unless the context clearly dictates otherwise.

Overall Configuration

FIG. **1** is a configuration diagram of a vehicle M in which a driving assistance device **100** of an embodiment is mounted. The vehicle M is, for example, a vehicle such as a two-wheeled vehicle, a three-wheeled vehicle, or a four-wheeled vehicle, and a drive source thereof is an internal combustion engine such as a diesel engine or a gasoline engine, an electric motor, or a combination thereof. The electric motor operates using electric power generated by a power generator connected to the internal combustion engine or electric power that is supplied when a secondary battery or a fuel cell is discharged.

For example, the vehicle M includes a camera **10**, a radar device **12**, a light detection and ranging (LIDAR) sensor **14**, an object recognition device **16**, a human machine interface (HMI) **30**, a vehicle sensor **40**, a navigation device **50**, driving operation elements **80**, a driving assistance device **100**, a travel driving force output device **200**, a brake device **210**, and a steering device **220**. Such devices and equipment are connected to each other by a multiplex communication line such as a controller area network (CAN) communication line, a serial communication line, or a wireless communication network. Also, the configuration shown in FIG. **1** is merely an example and some of the components may be omitted or other components may be further added.

For example, the camera **10** is a digital camera using a solid-state imaging element such as a charge-coupled device (CCD) or a complementary metal oxide semiconductor (CMOS). The camera **10** is attached to any location on the vehicle (hereinafter, the vehicle M) in which the vehicle system **1** is mounted. When the view in front of the vehicle M is imaged, the camera **10** is attached to an upper part of a front windshield, a rear surface of a rearview mirror, or the like. For example, the camera **10** periodically and iteratively images the surroundings of the vehicle M. The camera **10** may be a stereo camera.

The radar device **12** radiates radio waves such as millimeter waves around the vehicle M and detects at least a position (a distance to and a direction) of an object by detecting radio waves (reflected waves) reflected by the object. The radar device **12** is attached to any location on the vehicle M. The radar device **12** may detect a position and speed of the object in a frequency modulated continuous wave (FM-CW) scheme.

The LIDAR sensor **14** radiates light (or electromagnetic waves of a wavelength close to an optical wavelength) to the vicinity of the vehicle M and measures scattered light. The LIDAR sensor **14** detects a distance to an object on the basis of a time period from light emission to light reception. The radiated light is, for example, pulsed laser light. The LIDAR sensor **14** is attached to any location on the vehicle M.

The object recognition device **16** performs a sensor fusion process for detection results from some or all of the camera **10**, the radar device **12**, and the LIDAR sensor **14** to recognize a position, a type, a speed, and the like of an object. The object recognition device **16** outputs recognition results to the driving assistance device **100**. The object recognition device **16** may output detection results of the camera **10**, the radar device **12**, and the LIDAR sensor **14** to the driving assistance device **100** as they are. The object recognition device **16** may be omitted from the vehicle system **1**. Some or all of the camera **10**, the radar device **12**, the LIDAR sensor **14**, and the object recognition device **16** are an example of a "detection device."

The HMI **30** provides an occupant of the vehicle M with various types of information and receives an input operation from the occupant. The HMI **30** includes various types of display devices, a speaker, a buzzer, a vibration generation device (a vibrator), a touch panel, a switch, a key, and the like.

The vehicle sensor **40** includes a vehicle speed sensor configured to detect the speed of the vehicle M, an acceleration sensor configured to detect acceleration, a yaw rate sensor configured to detect an angular speed around a vertical axis, a direction sensor configured to detect a direction of the vehicle M, and the like.

The navigation device **50** has, for example, a global navigation satellite system (GNSS) receiver, a guidance controller, a storage storing map information, and the like. The GNSS receiver identifies a position of the vehicle M on the basis of signals received from GNSS satellites. A position of the vehicle M may be identified or corrected by an inertial navigation system (INS) using an output of the vehicle sensor **40**. For example, the guidance controller decides on a route from the position of the vehicle M identified by the GNSS receiver (or any input position) to a destination input by the occupant with reference to the map information and causes the HMI **30** to output guidance information so that the vehicle M travels along a path. The map information is, for example, information in which a road shape is expressed by a link indicating a road and nodes connected by the link. The map information may include curvature of a road, point of interest (POI) information, and the like. The navigation device **50** may transmit a current position and a destination of the vehicle M to a navigation server via the communication device and acquire a route from the navigation server.

The driving operation element **80** includes, for example, an accelerator pedal, a brake pedal, a steering wheel, a shift lever, and other operation elements. A sensor for detecting an operation amount or the presence or absence of an operation is attached to the driving operation element **80** and a detection result thereof is output to some or all of the travel driving force output device **200**, the brake device **210**, and the steering device **220**.

The travel driving force output device **200** outputs a travel driving force (torque) for enabling the vehicle to travel to driving wheels. For example, the travel driving force output device **200** includes a combination of an internal combustion engine, an electric motor, a transmission, and the like, and an electronic control unit (ECU) that controls the internal combustion engine, the electric motor, the transmission, and the like. The ECU controls the above-described components in accordance with information input from the driving assistance device **100** or information input from the driving operation element **80**.

For example, the brake device **210** includes a brake caliper, a cylinder configured to transfer hydraulic pressure to the brake caliper, an electric motor configured to generate hydraulic pressure in the cylinder, and an ECU. The ECU controls the electric motor in accordance with the information input from the driving assistance device **100** or the information input from the driving operation element **80** so that brake torque according to a braking operation is output to each wheel. The brake device **210** may include a mechanism configured to transfer the hydraulic pressure generated according to an operation on the brake pedal included in the driving operation elements **80** to the cylinder via a master cylinder as a backup. The brake device **210** is not limited to the above-described configuration and may be an electronically controlled hydraulic brake device configured to control an actuator in accordance with information input from the driving assistance device **100** and transfer the hydraulic pressure of the master cylinder to the cylinder.

For example, the steering device **220** includes a steering ECU and an electric motor. For example, the electric motor changes a direction of steerable wheels by applying a force to a rack and pinion mechanism. The steering ECU drives the electric motor in accordance with the information input from the driving assistance device **100** or the information input from the driving operation element **80** to change the direction of the steerable wheels.

Driving Assistance Device

The driving assistance device **100** includes, for example, a braking controller **110**, a steering-based avoidance controller **120**, and a second preliminary operation controller **130**. The braking controller **110** includes a first preliminary operation controller **112** and the second preliminary operation controller **130** includes a steering-based avoidance possibility determiner **132**. Each of these functional components is implemented, for example, by a hardware processor such as a central processing unit (CPU) executing a program (software). Some or all of the above components may be implemented by hardware (including a circuit; circuitry) such as a large-scale integration (LSI) circuit, an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), or a graphics processing unit (GPU) or may be implemented by software and hardware in cooperation. The program may be pre-stored in a storage device (a storage device including a non-transitory storage medium) such as a hard disk drive (HDD) or a flash memory of the driving assistance device **100** or may be stored in a removable storage medium such as a digital video disc (DVD) or a compact disc (CD)-read-only memory (ROM) and installed in the HDD or the flash memory of the driving assistance device **100** when the storage medium (the non-transitory storage medium) is mounted in a drive device. Functional components of some or all of the braking controller **110**, the steering-based avoidance controller **120**, and the second preliminary operation controller **130** may be integrated.

Setting is performed inside of the travel driving force output device **200**, the brake device **210**, and the steering device **220** so that instructions from the driving assistance device **100** to the travel driving force output device **200**, the brake device **210**, and the steering device **220** are issued with preference over a detection result from the driving operation element **80**. In relation to braking, if a braking force based on an operation amount of the brake pedal is larger than that in the instruction from the driving assistance device **100**, setting may be performed so that the braking operation is preferentially executed. As a mechanism for preferentially issuing an instruction from the driving assistance device **100**, the communication priority in the in-vehicle LAN may be used.

FIG. **2** is a diagram showing an overview of a function of the driving assistance device **100**. Hereinafter, each part of the driving assistance device **100** will be described with reference to FIG. **2** and FIG. **1**. In FIG. **2**, the vehicle M is traveling on a three-lane road and is in a lane L**2** in the center thereof. D_M denotes a traveling direction of the vehicle M.

The braking controller **110** instructs the brake device **210** and/or the travel driving force output device **200** to decelerate and stop the vehicle M when a degree of proximity between a target object TO among objects and the vehicle M satisfies a first condition with reference to an output of the detection device (described above) that detects that an object is located in front of the vehicle M. The target object TO is an object located on the same travel path as the vehicle M and on the traveling direction side of the vehicle M and is an object with which the vehicle M should avoid a collision, rather than objects that the vehicle M can pass over such as manholes. The braking controller **110** extracts such an object and sets the extracted object as the target object TO. In the example of FIG. **2**, another vehicle at the rear end of the conventional example is set as the target object TO. The travel path is, for example, a lane, but may be a virtual lane virtually set by the vehicle M on a road surface on which there is no road marking. The same is also true for the following description.

The "degree of proximity" is represented by various types of indicator values that indicate the degree of proximity between objects. For example, the "degree of proximity" is time to collision (TTC), which is an indicator value obtained by dividing a distance by a relative speed (positive in a direction in which objects approach each other). When the relative speed is negative (in a direction in which objects move away from each other), the TTC is provisionally set to infinity. The TTC is an indicator value indicating that the "degree of proximity" increases as the value decreases. The fact that the "first condition" is satisfied indicates, for example, that the TTC is less than a first threshold Th**1**. The first threshold Th**1** is, for example, a value of about 1.1 to 1.9 [sec]. Instead of the TTC, an indicator value having a similar property thereto, for example, a headway time, a distance, or another indicator value, may be used as the "degree of proximity" The TTC adjusted in consideration of acceleration and jerk may be used as the "degree of proximity" In the following description, it is assumed that the "degree of proximity" is the TTC.

When the TTC is less than the first threshold Th**1**, for example, the braking controller **110** instructs the brake device **210** and/or the travel driving force output device **200** to output a braking force for decelerating the vehicle M at first deceleration B**1**. The first deceleration B**1** is, for example, a deceleration of about 0.1 to 0.9 [G] (close to 1). Thereby, the braking controller **110** causes the vehicle M to quickly decelerate and stop and avoids a collision with the target object TO. The ECU of the brake device **210** or the travel driving force output device **200** has a function of obtaining a brake output, a regeneration control amount, an engine brake amount, or the like from instruction-specific deceleration. The ECU decides on each control amount on the basis of the instruction-specific deceleration and the speed of the vehicle M. This is well-known technology and detailed description thereof will be omitted.

The operation of the first preliminary operation controller **112** will be described below and the steering-based avoidance controller **120** will be described first.

FIG. **3** is a diagram showing an example of an operation scene of the steering-based avoidance controller **120**. The steering-based avoidance controller **120** determines whether or not there is a space where the vehicle M is able to travel in a travel path (for example, a lane L**1** or L**2**) on a lateral side of the target object TO when it is determined that it is difficult for the braking controller **110** to stop the vehicle M in front of the target object TO, generates an avoidance trajectory ET when it is determined that there is a space, and issues an instruction to the steering device **220** so that the vehicle M travels along the avoidance trajectory ET (steering-based avoidance). For example, the steering-based avoidance controller **120** determines whether or not an object is located in lateral side areas extending slightly in front of and behind the target vehicle on both lateral sides of the target vehicle TO, such as areas A**2**L and A**2**R shown in FIG. **3**, and determines that there is a space where the vehicle M is able to travel in a travel path on a lateral side of the target object TO when there is no object. The determination of whether or not it is difficult for the braking controller **110** to stop the vehicle M in front of the target object TO may be made by the braking controller **110**, or may be made by the steering-based avoidance controller **120**. The steering-based avoidance controller **120** may also recognize a boundary of a travel path by, for example, recognizing a white line or a road shoulder of a camera image, and determine that an object is located in an area when neither of the possible travel areas A**2**L and A**2**R is present, for example, when neither of the lanes L**1** and L**3** is present.

Steering-based avoidance is performed in a situation in which a sudden change in the surrounding environment of the vehicle has occurred such as a situation in which a target object TO decelerates unexpectedly or an object different from a recognized target object TO intervenes between the vehicle M and the target object TO and is set as a new target vehicle TO. In this situation, there is a possibility that a countermeasure cannot be taken at deceleration calculated in advance so that the vehicle stops in front of the target vehicle TO, but it is possible to increase a probability that sudden changes in the surrounding environment of the vehicle can be coped with by providing a steering-based avoidance function.

Preliminary Operation

The processes of the first preliminary operation controller **112** and the second preliminary operation controller **130** will be described below. FIG. **4** is a diagram for describing a preliminary operation.

When a degree of proximity between a target object TO and the vehicle M satisfies a second condition (for example, when the TTC is less than a second threshold Th**2**), the first preliminary operation controller **112** performs a first preliminary operation for notifying a driver of the vehicle M of the presence of the target object TO. The first preliminary operation is, for example, an operation of instructing the brake device **210** and/or the travel driving force output device **200** to output a braking force for decelerating the vehicle M at second deceleration B**2** from the time when the TTC is less than the second threshold Th**2** to the time when the TTC is less than the first threshold Th**1**. The second deceleration B**2** is deceleration less than the first deceleration B**1** (or close to zero). The second threshold Th**2** is a value larger than the first threshold Th**1**. Accordingly, the first condition is a condition that is satisfied when the degree of proximity is higher than that of the second condition.

When it is determined that the degree of proximity between the target object TO and the vehicle M satisfies a third condition (for example, the TTC is less than a third threshold Th**3**) and there is no available travel space in both travel paths on lateral sides of the target object TO after the vehicle M avoids a collision with the target object TO in steering at the time point when the third condition is satisfied, the second preliminary operation controller **130** performs a second preliminary operation of notifying the driver of the vehicle M of the presence of the target object TO. A determination related to the available travel space is made by the steering-based avoidance possibility determiner **132**. The third threshold Th**3** is a value larger than the second threshold Th**2**. Accordingly, the second condition is a condition that is satisfied when the degree of proximity is higher than that of the third condition.

For example, the steering-based avoidance possibility determiner **132** determines whether or not an object is located within lateral side areas extending slightly in front of and behind the target vehicle on both lateral sides of the target vehicle TO, such as areas A**1**L and A**1**R shown in FIG. **4**, for example, at a time point when the TTC is less than the third threshold Th**3**, and determines that there is a space where the vehicle M is able to travel in a travel path on the lateral side of the target object TO when there is no object. The areas A**1**L and A**1**R are set to be larger than the areas A**2**L and A**2**R, respectively, for example, in consideration of future uncertain factors. Like the steering-based avoidance controller **120**, the steering-based avoidance possibility determiner **132** may also recognize the boundary of the travel path by recognizing a white line and a road shoulder in a camera image and determine that an object is located in the area when neither of the possible travel areas A**1**L and A**1**R is present, for example, when neither of the lanes L**1** and L**3** is present. In the example of FIG. **4**, because there is no object in the area A**1**R, the steering-based avoidance possibility determiner **132** determines that there is a space where the vehicle M is able to travel in a travel path on the lateral side of the target object TO.

The second preliminary operation is, for example, an operation of instructing the brake device **210** and/or the travel driving force output device **200** to output a braking force for decelerating the vehicle M at third deceleration B**3** from the time when the TTC is less than the third threshold

Th3 to the time when the TTC is less than the first threshold Th1 and subsequently instructing the brake device 210 and/or the travel driving force output device 200 to output a braking force for decelerating the vehicle M at fourth deceleration B4. The third deceleration B3 is, for example, deceleration less than the second deceleration B2 (or close to zero), and the fourth deceleration B4 is deceleration greater than or substantially equal to the second deceleration and less than the first deceleration B1. A timing when the deceleration is switched from the third deceleration B3 to the fourth deceleration B4 may be set arbitrarily. For example, when a degree of proximity between the target object TO and the vehicle M satisfies a 1.5^{th} condition (for example, when the TTC is less than a threshold Th1.5), the deceleration is switched from the third deceleration B3 to the fourth deceleration B4. The 1.5^{th} condition is a condition that is satisfied when a degree of proximity is higher than that of the second condition. When the second condition is satisfied, the deceleration may be switched from the third deceleration B3 to the fourth deceleration B4.

Thus, a start timing of the second preliminary operation is earlier than that of the first preliminary operation and the second preliminary operation is performed in multiple steps. As described above, in a situation in which steering-based avoidance is possible, a probability that any sudden change in the surrounding environment of the vehicle can be coped with quickly becomes high and a degree of control margin becomes relatively high. On the other hand, because it is difficult to execute a steering-based avoidance function even if the steering-based avoidance function is provided when there is no avoidance space on the lateral side of the target object, a degree of control margin is no different from that of a vehicle that performs only an automated stop operation. That is, it is preferable to warn the driver of the vehicle M more quickly and effectively in a situation in which steering-based avoidance is difficult than in a situation in which steering-based avoidance is possible. According to the present embodiment, a start timing of the second preliminary operation is earlier than that of the first preliminary operation and the second preliminary operation is performed in multiple steps, and therefore it is possible to perform an appropriate preliminary operation corresponding to the surrounding situation of the target object.

FIG. 5 is a flowchart showing an example of a flow of a process executed by the driving assistance device 100.

First, the braking controller 110 identifies a target object TO (step S1). Subsequently, the second preliminary operation controller 130 determines whether or not TTC between the vehicle M and the target object TO is less than the third threshold Th3 (step S2). When the TTC between the vehicle M and the target object TO is greater than or equal to the third threshold Th3, the process returns to step S1.

When it is determined that the TTC between the vehicle M and the target object TO is less than the third threshold Th3, the steering-based avoidance possibility determiner 132 of the second preliminary operation controller 130 determines whether or not there is a space where the vehicle M is able to travel in a travel path on a lateral side of the target object TO (step S3).

When it is determined that there is no space where the vehicle M is able to travel in the travel path on the lateral side of the target object TO, the second preliminary operation controller 130 executes the second preliminary operation (step S4). Subsequently, the second preliminary operation controller 130 determines whether or not the TTC between the vehicle M and the target object TO has increased to a value greater than or equal to the third threshold Th3 (step S5). When it is determined that the TTC between the vehicle M and the target object TO has increased to a value greater than or equal to the third threshold Th3, the process returns to step S1.

When it is determined that the TTC between the vehicle M and the target object TO has not increased to a value greater than or equal to the third threshold Th3, the braking controller 110 determines whether or not the TTC between the vehicle M and the target object TO is less than the first threshold Th1 (step S6). When it is determined that the TTC between the vehicle M and the target object TO is greater than or equal to the first threshold Th1, the process returns to step S3. When an affirmative determination has been obtained in step S3, the second preliminary operation is stopped and the processing from step S8 is executed. When it is determined that the TTC between the vehicle M and the target object TO is less than the first threshold Th1, the braking controller 110 causes the vehicle M to decelerate and stop by causing the brake device 210 and/or the travel driving force output device 200 to output a braking force for decelerating the vehicle M at the first deceleration B1 (step S7). At this time, as described above, in place of (or in addition to) decelerating and stopping the vehicle M, steering-based avoidance may be performed.

When an affirmative determination has been obtained in step S3, i.e., when the TTC between the vehicle M and the target object TO is less than the third threshold Th3, and there is a space where the vehicle M is able to travel in the travel path on the lateral side of the target object TO, the first preliminary operation controller 112 of the braking controller 110 determines whether or not the TTC between the vehicle M and the target object TO is less than the second threshold Th2 (step S8). When it is determined that the TTC between the vehicle M and the target object TO is greater than or equal to the second threshold Th2, the process returns to step S1.

When it is determined that the TTC between the vehicle M and the target object TO is less than the second threshold Th2, the first preliminary operation controller 112 executes the first preliminary operation (step S9). Subsequently, the first preliminary operation controller 112 determines whether or not the TTC between the vehicle M and the target object TO has increased to a value greater than or equal to the second threshold Th2 (step S10). When it is determined that the TTC between the vehicle M and the target object TO has increased to a value greater than or equal to the second threshold Th2, the process returns to step S1.

When it is determined that the TTC between the vehicle M and the target object TO has not increased to a value greater than or equal to the second threshold Th2, the braking controller 110 determines whether or not the TTC between the vehicle M and the target object TO is less than the first threshold Th1 (step S11). When it is determined that the TTC between the vehicle M and the target object TO is greater than or equal to the first threshold Th1, the process returns to step S3. When a negative determination has been obtained in step S3, the first preliminary operation is stopped and the processing from step S4 is executed. When it is determined that the TTC between the vehicle M and the target object TO is less than the first threshold Th1, the braking controller 110 causes the brake device 210 and/or the travel driving force output device 200 to output the first deceleration B1 and causes the vehicle M to decelerate and stop (step S7).

According to the above-described embodiment, when the degree of proximity between the target object TO and the vehicle M satisfies the third condition and it is determined, at a time point when the third condition is satisfied, that there is no space where the vehicle M is able to travel in both travel paths on lateral sides of the target object TO after the vehicle M avoids a collision with the target object TO in steering, it is possible to perform an appropriate preliminary operation corresponding to a surrounding situation of the target object TO by performing the second preliminary operation started at an earlier timing than the first preliminary operation.

In the above-described embodiment, in either of the first preliminary operation and the second preliminary operation, a display process, a sound output process, or a vibration output process or the like as an alert may be performed instead of the output of the braking force. In this case, as an example in which the second preliminary operation is performed in multiple steps, instead of outputting the braking force stepwise while changing the degree of deceleration as described above, a process of differentiating a degree of attention (contrast, brightness, color, or the like) between an initial display screen and second and subsequent display screens, a process of differentiating content or a volume between an initial sound output and second and subsequent sound outputs, a process of increasing second and subsequent vibration outputs as compared with the first vibration output, or the like may be provided.

In the above-described embodiment, when the branch road to the destination set in the navigation device **50** is located on the left or right side of the lane in which the vehicle M is traveling, the lane change may be forcedly made during the preliminary operation. Thus, consequently, it is possible to move the vehicle M in a direction closer to the destination and guide the vehicle M in a state in which the object serving as the target object is not near the vehicle M.

Although an example in which a process is executed without taking into account a type of target object has been described above, a process may be executed in consideration of a type of target object as described below. Some of the first preliminary operation controller **112**, the second preliminary operation controller **130**, and the steering-based avoidance controller **120** change control content in accordance with the type of target object. For example, some of the first preliminary operation controller **112**, the second preliminary operation controller **130**, and the steering-based avoidance controller **120** change control content in accordance with whether the target object is a four-wheeled vehicle or a two-wheeled vehicle. The two-wheeled vehicle is, for example, a motorcycle, a light vehicle, or the like.

The fact that the control content is changed indicates, for example, that the timings of the first preliminary operation, the second preliminary operation, and the steering-based avoidance are changed or that a control amount of the operation or the steering is changed in accordance with the target object. For example, when the target object is the two-wheeled vehicle, the driving assistance device **100** suppresses the proximity of the vehicle to the two-wheeled vehicle by actively operating the brake device of the vehicle M (or by actively performing the first preliminary operation or the second preliminary operation). The driving assistance device **100** suppresses the proximity of the vehicle to the two-wheeled vehicle by more actively operating the brake device of the vehicle M when the target object is the two-wheeled vehicle than when the target object is the four-wheeled vehicle. For example, a timing at which the braking operation is performed (a timing of the first preliminary operation or the second preliminary operation) when the target object is the two-wheeled vehicle is earlier than that when the target object is the four-wheeled vehicle.

The steering-based avoidance controller **120** suppresses the steering-based avoidance operation when the target object is the two-wheeled vehicle as compared with when the target object is the four-wheeled vehicle. The steering-based avoidance controller **120** delays a timing of steering-based avoidance when the target object is the two-wheeled vehicle as compared with when the target object is the four-wheeled vehicle. A change example of control content will be described below.

Change Example (1) of Control Content

FIG. **6** is a diagram (1) for describing a change example of control content. In a change example (1), when the target object is a two-wheeled vehicle, the timing of the second preliminary operation is advanced. Differences from FIG. **4** will be described.

The second preliminary operation controller **130** performs the second preliminary operation at a timing after a fourth condition is satisfied (for example, when the TTC is less than the third threshold Th**3**) when the degree of proximity between the target object and the vehicle M satisfies the fourth condition, it is determined, at the time point when the fourth condition is satisfied, that there is no space where the vehicle M is able to travel in both travel paths on lateral sides of the target object after the vehicle M avoids the collision with the target object in the steering, and the target object is the four-wheeled vehicle. The second preliminary operation controller **130** performs the second preliminary operation at a timing in which the fourth condition is satisfied when the degree of proximity between the target object and the vehicle M satisfies the fourth condition (for example, the TTC is less than a fourth threshold Th**4**), it is determined, at the time point when the fourth condition is satisfied, that there is no space where the vehicle M is able to travel in both travel paths on the lateral sides of the target object after the vehicle M avoids the collision with the target object in the steering, and the target object is the two-wheeled vehicle. The third condition is a condition that is satisfied when the degree of proximity is higher than that of the fourth condition.

In the example of FIG. **6**, there is no space where the vehicle M is able to travel in both travel paths on the lateral sides of the target object after the vehicle M avoids the collision with the target object in the steering. In this case, the second preliminary operation controller **130** performs the second preliminary operation if the third condition is satisfied as described above when the target object is the four-wheeled vehicle, but performs the second preliminary operation if the fourth condition is satisfied when the target object is the two-wheeled vehicle. Subsequently, the second preliminary operation controller **130** switches the deceleration from the third deceleration B**3** to the fourth deceleration B**4** at a timing between the timing when the second condition is satisfied and the timing when the first condition is satisfied (for example, a timing when the TTC is less than the threshold Th**1.5**). The second preliminary operation controller **130** may switch the deceleration from the third deceleration B**3** to the fourth deceleration B**4** at a timing earlier than the timing when the TTC is less than the threshold Th**1.5**.

For example, it is desirable to allow the driver of the vehicle M to recognize that the two-wheeled vehicle is approaching the vehicle M with time to spare when the target object is the two-wheeled vehicle as compared with when the target object is the four-wheeled vehicle. For example, when a vehicle is approaching from behind, the approach of the vehicle is more noticeable in the case of riding the two-wheeled vehicle than in the case of riding the four-wheeled vehicle. Thus, when the target object is the two-wheeled vehicle, it is desirable to suppress the degree of proximity of the rear vehicle earlier than when the target object is the four-wheeled vehicle. Furthermore, it is desirable to suppress proximity more reliably because a larger impact is given if the vehicle comes into contact with the two-wheeled vehicle than if the vehicle comes into contact with the four-wheeled vehicle. Thus, as described above, the second preliminary operation controller **130** advances the timing of the second preliminary operation when the target object is the two-wheeled vehicle. The second preliminary operation controller **130** can perform an appropriate preliminary operation corresponding to the surrounding situation of the target object.

Change Example (2) of Control Content

FIG. **7** is a diagram (2) for describing a change example of control content. In the change example (2), the second preliminary operation is performed when the target object is a two-wheeled vehicle even if a condition for performing the first preliminary operation is satisfied. Differences from FIG. **4** will be described.

The second preliminary operation controller **130** does not perform the second preliminary operation at a timing when the third condition is satisfied when the degree of proximity between the target object and the vehicle M satisfies the third condition, it is determined, at a time point when the third condition is satisfied, that there is a space where the vehicle M is able to travel in one of the travel paths on the lateral sides of the target object after the vehicle M avoids the collision with the target object in the steering, and the target object is the four-wheeled vehicle. In this case, the first preliminary operation controller **112** performs the first preliminary operation when the degree of proximity between the four-wheel vehicle and the vehicle satisfies the second condition.

The second preliminary operation controller **130** performs the second preliminary operation at a timing when the third condition is satisfied when the degree of proximity between the target object and the vehicle M satisfies the third condition, it is determined, at a time point when the third condition is satisfied, that there is a space where the vehicle M is able to travel in one of the travel paths on the lateral sides of the target object after the vehicle M avoids the collision with the target object in the steering, and the target object is the two-wheeled vehicle. In other words, when there is a space where avoidance based on steering is possible in the travel paths on the lateral sides of the target object and the target object is the two-wheeled vehicle, the second preliminary operation is performed instead of the first preliminary operation.

In the example of FIG. **7**, there is a space where the vehicle M is able to travel in one of the travel paths on the lateral sides of the target object after the vehicle M avoids the collision with the target object in the steering. In this case, when the target object is the two-wheeled vehicle and the third condition is satisfied, the second preliminary operation controller **130** performs the second preliminary operation. Subsequently, the second preliminary operation controller **130** switches the deceleration from the third deceleration B**3** to the fourth deceleration B**4** at a timing between the timing when the second condition is satisfied and the timing when the first condition is satisfied (for example, the timing when the TTC is less than the threshold Th**1.5**). The second preliminary operation controller **130** may switch the deceleration from the third deceleration B**3** to the fourth deceleration B**4** at a timing earlier than the timing when the TTC is less than the threshold Th**1.5**.

As described above, when the target object is the two-wheeled vehicle, the second preliminary operation is performed instead of the first preliminary operation. The second preliminary operation controller **130** can perform an appropriate preliminary operation corresponding to the surrounding situation of the target object.

Change Example (3) of Control Content

FIG. **8** is a diagram (3) for describing a change example of control content. A timing at which the second preliminary operation is performed in the change example (3) is earlier than that in the change example (2). Differences from FIG. **7** will be described.

The second preliminary operation controller **130** does not perform the second preliminary operation at a timing when a fifth condition is satisfied when the degree of proximity between the target object and the vehicle M satisfies the fifth condition (for example, the degree of proximity between the target object and the vehicle M is less than a fifth threshold Th**5**), it is determined, at a time point when the fifth condition is satisfied, that there is a space where the vehicle M is able to travel in one of the travel paths on the lateral sides of the target object after the vehicle M avoids the collision with the target object in the steering, and the target object is the four-wheeled vehicle. In this case, the first preliminary operation controller **112** performs the first preliminary operation when the degree of proximity between the four-wheeled vehicle and the vehicle M satisfies the second condition.

The second preliminary operation controller **130** performs the second preliminary operation at a timing when the fifth condition is satisfied when the degree of proximity between the target object and the vehicle M satisfies the fifth condition, it is determined, at a time point when the fifth condition is satisfied, that there is a space where the vehicle M is able to travel in one of the travel paths on the lateral sides of the target object after the vehicle M avoids the collision with the target object in the steering, and the target object is the two-wheeled vehicle. The second condition is a condition that is satisfied when the degree of proximity is higher than that of the fifth condition and the third condition is a condition that is satisfied when the degree of proximity is higher than that of the fifth condition. The fifth condition may be a condition that is satisfied when the degree of proximity is higher than that of the fourth condition.

In the example of FIG. **8**, there is a space where the vehicle M is able to travel in one of the travel paths on the lateral sides of the target object after the vehicle M avoids the collision with the target object in the steering. In this case, when the target object is the two-wheeled vehicle and the fifth condition is satisfied (or when the TTC is the fifth threshold Th**5**), the second preliminary operation controller **130** performs the second preliminary operation. Subsequently, the second preliminary operation controller **130** switches the deceleration from the third deceleration B**3** to the fourth deceleration B**4** at a timing between the timing when the second condition is satisfied and the timing when the first condition is satisfied (for example, a timing when the TTC is less than the threshold Th**1.5**). The second preliminary operation controller **130** may switch the deceleration from the third deceleration B3 to the fourth deceleration B4 at a timing earlier than the timing when the TTC is less than the threshold Th1.5.

As described above, when the target object is the two-wheeled vehicle and the fifth condition is satisfied, the second preliminary operation is performed at an earlier timing instead of the first preliminary operation. The second preliminary operation controller **130** can perform an appropriate preliminary operation corresponding to the surrounding situation of the target object.

Change Example (4) of Control Content

FIG. **9** is a diagram (4) for describing a change example of control content. In the change example (4), a timing at which the first preliminary operation is performed is advanced. Differences from FIG. **4** will be described.

The first preliminary operation controller **112** does not perform the first preliminary operation at a timing when a sixth condition is satisfied when the degree of proximity between the target object and the vehicle M satisfies the sixth condition (for example, the TTC is less than a threshold Th**2** #), it is determined, at a time point when the sixth condition is satisfied, that there is a space where the vehicle M is able to travel in one of the travel paths on the lateral sides of the target object after the vehicle M avoids the collision with the target object in the steering, and the target object is the four-wheeled vehicle, and performs the first preliminary operation when the degree of proximity between the four-wheeled vehicle and the vehicle M satisfies the second condition. The second condition is a condition that is satisfied when the degree of proximity is higher than that of the sixth condition. The sixth condition may be, for example, a condition that is satisfied when the degree of proximity is higher than that of the third condition.

The first preliminary operation controller **112** performs the first preliminary operation at a timing when the sixth condition is satisfied when the degree of proximity between the target object and the vehicle M satisfies the sixth condition, it is determined, at a time point when the sixth condition is satisfied, that there is a space where the vehicle M is able to travel in one of the travel paths on the lateral sides of the target object after the vehicle M avoids the collision with the target object in the steering, and the target object is the two-wheeled vehicle.

In the example of FIG. **9**, there is a space where the vehicle M is able to travel in one of the travel paths on the lateral sides of the target object after the vehicle M avoids the collision with the target object in the steering. In this case, when the target object is the two-wheeled vehicle and the sixth condition is satisfied (when the TTC is the threshold Th**2** #), the first preliminary operation controller **112** performs the first preliminary operation.

As described above, when the target object is the two-wheeled vehicle and the sixth condition is satisfied, a timing when the first preliminary operation is performed is advanced. The first preliminary operation controller **112** can perform an appropriate preliminary operation corresponding to the surrounding situation of the target object.

Change Example (5) of Control Content

FIG. **10** is a diagram (5) for describing a change example of control content. In change example (5), steering control content is changed. FIG. **10** is a diagram showing an example of behavior of the vehicle M when the target object is a four-wheeled vehicle. FIG. **11** is a diagram showing an example of behavior of the vehicle M when the target object is a two-wheeled vehicle. Although an example in which the first preliminary operation is performed will be described in FIGS. **10** and **11**, the steering is similarly controlled even if the second preliminary operation is performed.

The steering-based avoidance controller **120** issues an instruction to the steering device of the vehicle M so that control for avoiding a collision with the target object in steering is suppressed when the target object is the two-wheeled vehicle as compared with when the target object is the four-wheeled vehicle. The suppression is, for example, to suppress the proximity to the target object by giving priority to braking over avoidance based on steering. The suppression is, for example, to delay a timing of avoidance based on steering. In the suppression, steering-based avoidance may not be performed.

If the target object is the four-wheeled vehicle, as shown in FIG. **10**, when the distance between the vehicle M and the four-wheeled vehicle becomes a distance d (for example, when the degree of proximity reaches a first prescribed threshold), the steering-based avoidance controller **120** controls steering so that the vehicle M is moved to a lane L**3**.

If the target object is the two-wheeled vehicle, as shown in FIG. **11**, when the distance between the vehicle M and the two-wheeled vehicle becomes a distance d #(for example, when the degree of proximity reaches a second prescribed threshold), the steering-based avoidance controller **120** controls steering so that the vehicle M is moved to a lane L**3**. The distance d #is a distance shorter than the distance d. When the target object is the two-wheeled vehicle, the vehicle M may decelerate at higher deceleration (deceleration higher than the deceleration B**1**) than when the target object is the four-wheeled vehicle.

For example, four-wheeled vehicles cannot change lanes quickly. At the distance d, if the four-wheeled vehicle does not attempt to change lanes, even if the four-wheeled vehicle subsequently starts to change lanes, the lane change of the vehicle M is completed before the four-wheeled vehicle enters a lane that is a lane change destination. Thus, interference between the vehicle M and the four-wheeled vehicle does not occur in many cases.

For example, two-wheeled vehicles can quickly change lanes. At the distance d, even if the two-wheeled vehicle does not attempt to change lanes, if the two-wheeled vehicle subsequently starts to change lanes, the vehicle M may change lanes at the same timing when the two-wheeled vehicle changes lanes. The possibility of interference between the vehicle M and the two-wheeled vehicle cannot be denied. Thus, as shown in FIG. **11**, when the target object is a two-wheeled vehicle, the vehicle M changes lanes at the timing of the distance d #. If it is the timing of the distance d #, the vehicle M is expected to make a lane change to the lane L**3** and complete the lane change before the two-wheeled vehicle enters the lane L**3** even if the two-wheeled vehicle starts the lane change operation at the timing.

Thus, the steering-based avoidance controller **120** can perform an appropriate preliminary operation corresponding to a surrounding situation of a target object by suppressing control for avoiding a collision with the target object in steering when the target object is a two-wheeled vehicle as compared with when the target object is a four-wheeled vehicle.

As described above, some of the first preliminary operation controller **112**, the second preliminary operation controller **130**, and the steering-based avoidance controller **120** can perform an appropriate preliminary operation corresponding to the surrounding conditions of the target object by changing control content in accordance with a type of target object.

All the above-described processes may be implemented or some processes may be selectively implemented.

Although an example in which control content is changed in accordance with whether the target object is a four-wheeled vehicle or a two-wheeled vehicle has been described above, control content may be changed in accordance with whether the target object is a four-wheeled vehicle or another object in place of (or in addition to) the example. For example, in the case of the other object, control similar to that in the case of a two-wheeled vehicle may be performed.

The embodiment described above can be represented as follows.

A driving assistance device including:

a storage medium storing computer-readable instructions; and

a processor connected to the storage medium, the processor executing the computer-readable instructions to execute:

a referring process of referring to an output of a detection device for detecting that an object is located in front of a vehicle

one or both of a braking control process or a steering based avoidance control process;

the braking control process being a process that instructs a brake device of the vehicle to stop the vehicle when a degree of proximity between a target object among objects and the vehicle satisfies a first condition,

the steering-based avoidance control process being a process that instruct instruct a steering device of the vehicle to avoid a collision with the target object in steering,

a first preliminary operation control process of performing a first preliminary operation when the degree of proximity between the target object and the vehicle satisfies a second condition; and

a second preliminary operation control process of performing a second preliminary operation when the degree of proximity between the target object and the vehicle satisfies a third condition it is determined, at a time point when the third condition is satisfied, that there is no space where the vehicle is able to travel in both travel paths on lateral sides of the target object after the vehicle avoids the collision with the target object in the steering,

wherein the first condition is a condition that is satisfied when the degree of proximity is higher than that of the second condition,

wherein the second condition is a condition that is satisfied when the degree of proximity is higher than that of the third condition, and

wherein control content is changed in accordance with a type of the object in at least some of the steering-based avoidance control process, the first preliminary operation control process, and the second preliminary operation control process.

Although modes for carrying out the present invention have been described above using embodiments, the present invention is not limited to the embodiments and various modifications and substitutions can also be made without departing from the scope and spirit of the present invention.

What is claimed is:

1. A driving assistance device comprising:

a storage medium storing computer-readable instructions; and

at least one processor connected to the storage medium, the at least one processor executing the computer-readable instructions to:

execute a referring process of referring to an output of a detection device for detecting that an object is located in front of a vehicle;

execute one or both of a braking control process or a steering based avoidance control process;

the braking control process being a process that instructs a brake device of the vehicle to stop the vehicle when an indicator value that decreases as the vehicle approaches the object is less than a first threshold, the a steering-based avoidance control process being a process that instruct a steering device of the vehicle to avoid a collision with the object in steering,

execute a first preliminary operation control process of performing a first preliminary operation of notifying a driver of the vehicle that the object is present when the indicator value is less than a second threshold; and

execute a second preliminary operation control process of performing a second preliminary operation of notifying the driver of the vehicle that the object is present when the indicator value is less than a third threshold and it is determined, at a time point when the indicator value is less than the third threshold, that there is no travel path along which the vehicle is able to travel on both left and right sides of the object after the vehicle avoids the collision with the object in the steering,

wherein the first threshold is less than the second threshold and the second threshold is less than the third threshold,

wherein control content is changed in accordance with a type of the object in at least some of the steering-based avoidance control process, the first preliminary operation control process, and the second preliminary operation control process,

the first preliminary operation includes some or all of outputting braking force, displaying, outputting sound, and outputting vibration,

the second preliminary operation includes some or all of the outputting braking force, the displaying, the outputting sound, and the outputting vibration, and

the at least one processor executing the computer-readable instructions to perform (i) or (ii):

(i) when the type of the object is a two-wheeled vehicle, the timing of the first preliminary operation or the second preliminary operation is earlier than when the type of the object is a four-wheeled vehicle, or

(ii) the second preliminary operation is executed when the type of the object is the two-wheeled vehicle, and the second preliminary operation is not executed when the type of the object is the four-wheeled vehicle.

2. The driving assistance device according to claim **1**, wherein the at least one processor changes control content in accordance with whether the object is a four-wheeled vehicle or a two-wheeled vehicle in at least some of the steering-based avoidance control process, the first preliminary operation control process, and the second preliminary operation control process.

3. The driving assistance device according to claim **1**, wherein, when the object is a two-wheeled vehicle, the

driving assistance device suppresses proximity of the vehicle to the two-wheeled vehicle by actively operating the brake device of the vehicle.

4. The driving assistance device according to claim **1**, wherein the at least one processor suppresses the avoidance based on the steering in the steering-based avoidance control process when the object is a two-wheeled vehicle as compared with when the object is a four-wheeled vehicle.

5. The driving assistance device according to claim **1**,

wherein the at least one processor performs the second preliminary operation at a time point after the indicator value is less than a fourth threshold when the indicator value is less than the fourth threshold, it is determined, at a time point when the indicator value is less than the fourth threshold, that there is no travel path along which the vehicle is able to travel on both left and right sides of the object after the vehicle avoids the collision with the object in the steering, and the object is a four-wheeled vehicle,

wherein the at least one processor performs the second preliminary operation at a time point when the indicator value is less than the fourth threshold when the indicator value is less than the fourth threshold, it is determined, at the time point when the indicator value is less than the fourth threshold, that there is no travel path along which the vehicle is able to travel on both left and right sides of the object after the vehicle avoids the collision with the object in the steering, and the object is a two-wheeled vehicle, and

wherein the third threshold is less than the fourth threshold.

6. The driving assistance device according to claim **1**,

wherein the at least one processor does not perform the second preliminary operation at a time point when the indicator value between a four-wheeled vehicle and the vehicle is less than the third threshold when the indicator value is less than the third threshold, it is determined, at a time point when the indicator value is less than the third threshold, that there is a travel path along which the vehicle is able to travel on one of the left and right sides of the object after the vehicle avoids the collision with the object in the steering, and the object is the four-wheeled vehicle, and performs the first preliminary operation when the indicator value between the four-wheeled vehicle and the vehicle is less than the second threshold, and

wherein the at least one processor performs the second preliminary operation at a time point when the indicator value between a two-wheeled vehicle and the vehicle is less than the third threshold when the indicator value is less than the third threshold, it is determined, at the time point when the indicator value is less than the third threshold, that there is a travel path along which the vehicle is able to travel on one of the left and right sides of the object after the vehicle avoids the collision with the object in the steering, and the object is the two-wheeled vehicle.

7. The driving assistance device according to claim **1**,

wherein the at least one processor does not perform the second preliminary operation at a time point when the indicator value is less than a fifth threshold when the indicator value is less than the fifth threshold, it is determined, at the time point when the indicator value is less than the fifth threshold, that there is a travel path along which the vehicle is able to travel on one of the left and right sides of the object after the vehicle avoids the collision with the object in the steering, and the object is a four-wheeled vehicle, and performs the first preliminary operation when the indicator value between the four-wheeled vehicle and the vehicle is less than the second threshold,

wherein the at least one processor performs the second preliminary operation at a time point when the indicator value between a two-wheeled vehicle and the vehicle is less than the fifth threshold when the indicator value is less than the fifth threshold, it is determined, at the time point when the indicator value is less than the fifth threshold, that there is a travel path along which the vehicle is able to travel on one of the left and right sides of the object after the vehicle avoids the collision with the object in the steering, and the object is the two-wheeled vehicle,

wherein the second threshold is less than the fifth threshold, and

wherein the third threshold is less than the fifth threshold.

8. The driving assistance device according to claim **1**,

wherein the at least one processor does not perform the first preliminary operation at a time point when the indicator value between a four-wheeled vehicle and the vehicle is less than a sixth threshold when the indicator value is less than the sixth threshold, it is determined, at a time point when the indicator value is less than the sixth threshold, that there is a travel path along which the vehicle is able to travel on one of the left and right sides of the object after the vehicle avoids the collision with the object in the steering, and the object is the four-wheeled vehicle, and performs the first preliminary operation when the indicator value between the four-wheeled vehicle and the vehicle is less than the second threshold,

wherein the at least one processor performs the first preliminary operation at a time point when the indicator value between a two-wheeled vehicle and the vehicle is less than the sixth threshold when the indicator value is less than the sixth threshold, it is determined, at the time point when the indicator value is less than the sixth threshold, that there is a travel path along which the vehicle is able to travel on one of the left and right sides of the object after the vehicle avoids the collision with the object in the steering, and the object is the two-wheeled vehicle, and

wherein the second threshold is less than the sixth threshold.

9. A driving assistance method comprising:

executing, by a driving assistance device, referring process of referring to an output of a detection device for detecting that an object is located in front of a vehicle;

executing, by the driving assistance device, one or both of a braking control process of instructing a brake device of the vehicle to stop the vehicle when an indicator value that decreases as the vehicle approaches the object is less than a first threshold, and a steering-based avoidance control process of instructing a steering device of the vehicle to avoid a collision with the object in steering;

executing, by the driving assistance device, a first preliminary operation control process of performing a first preliminary operation of notifying a driver of the vehicle that the object is present when the indicator value is less than a second threshold; and

executing, by the driving assistance device, a second preliminary operation control process of performing a second preliminary operation of notifying the driver of the vehicle that the object is present when the indicator value is less than a third threshold and it is determined, at a time point when the indicator value is less than the third threshold, that there is no travel path along which the vehicle is able to travel on both left and right sides of the object after the vehicle avoids the collision with the object in the steering,

wherein the first threshold is less than the second threshold and the second threshold is less than the third threshold,

wherein control content is changed in accordance with a type of the object in at least some of the steering-based avoidance control process, the first preliminary operation control process, and the second preliminary operation control process,

the first preliminary operation includes some or all of outputting braking force, displaying, outputting sound, and outputting vibration,

the second preliminary operation includes some or all of the outputting braking force, the displaying, the outputting sound, and the outputting vibration, and

the driving assistance method further comprising (i) or (ii):

(i) when the type of the object is a two-wheeled vehicle, the timing of the first preliminary operation or the second preliminary operation is earlier than when the type of the object is a four-wheeled vehicle, or

(ii) the second preliminary operation is executed when the type of the object is the two-wheeled vehicle, and the second preliminary operation is not executed when the type of the object is the four-wheeled vehicle.

10. A non-transitory computer storage medium storing a program for causing a computer to:

execute a referring process of referring to an output of a detection device for detecting that an object is located in front of a vehicle;

execute one or both of a braking control process of referring to an output of a detection device for detecting that an object is located in front of a vehicle and instructing a brake device of the vehicle to stop the vehicle when an indicator value that decreases as the vehicle approaches the object is less than a first threshold, and a steering-based avoidance control process of instructing a steering device of the vehicle to avoid a collision with the object in steering;

execute a first preliminary operation control process of performing a first preliminary operation of notifying a driver of the vehicle that the object is present when the indicator value is less than a second threshold; and

execute a second preliminary operation control process of performing a second preliminary operation of notifying the driver of the vehicle that the object is present when the indicator value is less than a third threshold and it is determined, at a time point when the indicator value is less than the third threshold, that there is no travel path along which the vehicle is able to travel on both left and right sides of the object after the vehicle avoids the collision with the object in the steering,

wherein the first threshold is less than the second threshold and the second threshold is less than the third threshold,

wherein control content is changed in accordance with a type of the object in at least some of the steering-based avoidance control process, the first preliminary operation control process, and the second preliminary operation control process,

the first preliminary operation includes some or all of outputting braking force, displaying, outputting sound, and outputting vibration,

the second preliminary operation includes some or all of the outputting braking force, the displaying, the outputting sound, and the outputting vibration, and

the non-transitory computer storage medium storing the program for causing the computer to perform (i) or (ii):

(i) when the type of the object is a two-wheeled vehicle, the timing of the first preliminary operation or the second preliminary operation is earlier than when the type of the object is a four-wheeled vehicle, or

(ii) the second preliminary operation is executed when the type of the object is the two-wheeled vehicle, and the second preliminary operation is not executed when the type of the object is the four-wheeled vehicle.

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